BACHELOR OF COMPUTER SCIENCE
SCHOOL OF COMPUTER SCIENCE
BINA NUSANTARA UNIVERSITY
JAKARTA

ASSESSMENT FORM

Course: COMP6798 – Program Design Methods

Method of Assessment: Case Study and Project

Semester/Academic Year: 1/2022–2023

Name of Lecturer : ………………………

Date : ………………………

Class : ………………………

Topic : **Input-Process-Output, Analysis Modeling, Business Process and Functional Modeling**

Group Members:	1 ____________________ 2 ____________________ 3 ____________________ 4 ____________________ 5 ____________________ 6 ____________________ 7 ____________________ 8 ____________________

Student Outcomes:
(SO 2) Mampu merancang, mengimplementasikan, dan mengevaluasi solusi berbasis komputasi untuk memenuhi serangkaian persyaratan komputasi dalam konteks ilmu komputer.
Able to design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of computer science.

Learning Objectives:
(LObj 2.1) Mampu merancang solusi berbasis komputasi untuk memenuhi serangkaian persyaratan komputasi tertentu dalam konteks ilmu

komputer.
Able to design a computing-based solution to meet a given set of computing requirements in the context of computer science.

No.	Assessment Criteria	Weight	Excellent (85–100)	Good (75–84)	Average (65–74)	Poor (0–64)	Score	(Score × Weight)
1	Ability to identify requirements for programs	**25%**	Able to identify the input, process, and output of a program completely and correctly.	Able to identify the input, process, and output of a program in less-completely and less-correctly.	Able to identify the input, process, and output of a program in half-completely and half-correctly.	Identify the input, process, and output of a program incompletely and incorrectly.		
2	Ability to apply pseudocode	**25%**	The pseudocode written solves at least 90% of the programming problem.	The pseudocode written solves at least 75% of the programming problem.	The pseudocode written solves at least 60% of the programming problem.	The pseudocode written solves less than 60% of the programming problem.		
3	Ability to collect the requirements for a system	**25%**	All the necessary requirement of a system is collected.	Some of the requirement of a system is collected.	A few of the requirement of a system is collected.	The requirement of a system is not collected.		
4	Ability to design a system using UML	**25%**	Able to design at least 90% correct using UML.	Able to design at least 75% correct using UML.	Able to design at least 60% correct using UML.	Able to design less than 60% correct using UML.		
	Total Score: ∑(Score × Weight)							

Remarks:

ASSESSMENT METHOD

Instructions

- This assignment is group work consisting of 5–7 students.
- Given a case study, students are required to:
  - Identify program requirements using IPO charts.
  - Develop the algorithm using pseudocode.
- Conduct an interview or observation with any Binusian on campus to learn more about their daily tasks in the system unit where they are positioned, then:
  - Gather the functional and non-functional requirements of the system.
  - Draw a UML diagram(s) to describe the process.
- The assignment is documented in a single .pdf report which then submitted to BINUSMAYA.

Note for Lecturers:

- Notify students to form a group at the first meeting so they can start working on the assignment as soon as possible
- Lecturers may create the groups in the BINUSMAYA so that students are able to submit the report.
- The report is submitted before the final exam (a week before is preferable) by one student only as the representation of each group.

QUESTIONS

1. Case Study (50%)

Since the COVID-19 pandemic, BINUS University has implemented health protocols in its campus environment. One of them is monitoring the body temperature of visitors before entering the campus building. In this process, security staff monitor body temperature sensors at every building entrance. If the visitor's body temperature is less than or equal to 37.1 degrees Celsius, then the visitor has a normal body temperature, so they are welcome to enter the building. However, if a visitor is indicated to have Covid, namely with a body temperature exceeding 37.1 degrees Celsius, the security staff will carry out a quarantine protocol by conducting an on-site antigen test. At the end of the day, the data record of body temperature for each visitor will be stored in a file called **Todaysvisitors.dat**.

As a developer, you are asked to create a pseudocode for an algorithm that calculates the total number of visitors with a normal body temperature and a body temperature suspected of COVID-19. The total number of visitors can be calculated based on the data record from the **Todaysvisitors.dat** file, as shown in the following Table 1.

Briella	36.9
Ivory	37.3
Nadia	36.9
Uriel	37.1
Sebastian	36.8
Ivan	37.2
Abigail	37.0
Nathanael	37.1
Thalia	36.9
Wynter	36.8
Eric	37.3
Nina	36.9
Tiffany	37.2
Yara	36.9
Samuel	37.2
Ingrid	36.9
Xavier	37.0

Table 1: **Todaysvisitors.dat** file containing visitors’ name and their body temperatures.

Your tasks are:

a. Define the requirements of the program with the IPO Chart. You may refer to IPO Chart definition in *Tony Gaddis. (2019). Starting Out with Programming Logic and Design. 5th Edition. Chapter 2.2 Output, Input, and Variables.*

b. Write pseudocode for a program that can calculate the total number of visitors based on body temperature using the **Todaysvisitors.dat** file. The pseudocode should display the following output:

```
Today's Visitor Count by Body Temperature:

Normal Body Temperature: __
Body temperature suspected for COVID: __
Total visitors: __
```

2. Project (50%)

a. Conduct an interview or an observation on any Binusian (e.g., students, lecturer, employee) to learn more about the daily tasks and the process it takes to complete the tasks. You may ask about what they do and analyze how they collaborate with others in the system where they are positioned.

As a group of programmers, you are interested in developing an application that automates their tasks and processes. Before developing the application, you are required to gather requirements that the application may need. Define **five functional** and **two non-functional requirements** of the system for the application.

b. Identify the actor(s), scenario(s), and use cases (processes) and draw a use case diagram that describes the requirements specified in Question 2.a. If you use either **<<extend>> or <<include>>**, please provide a detailed explanation(s) about the reason. As guidance in creating a use case diagram, you may refer to *Alan Dennis, Barbara Haley Wixom, and David Tegarden. (2015). Systems Analysis and Design: An Object–Oriented Approach with UML. 5th Edition. Chapter 4: Business Process and Functional Modeling. Page 126–127.*

-- Good luck --